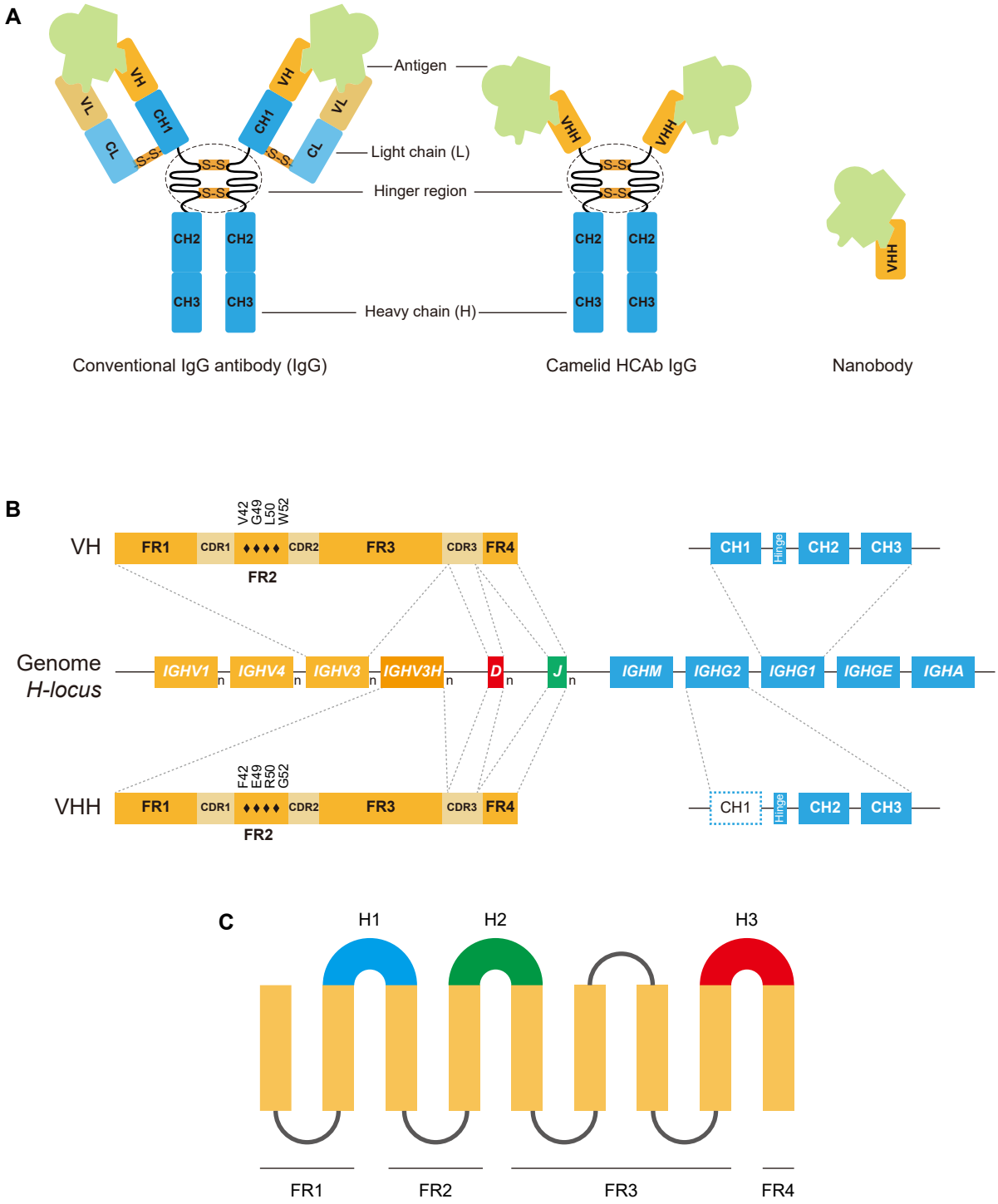
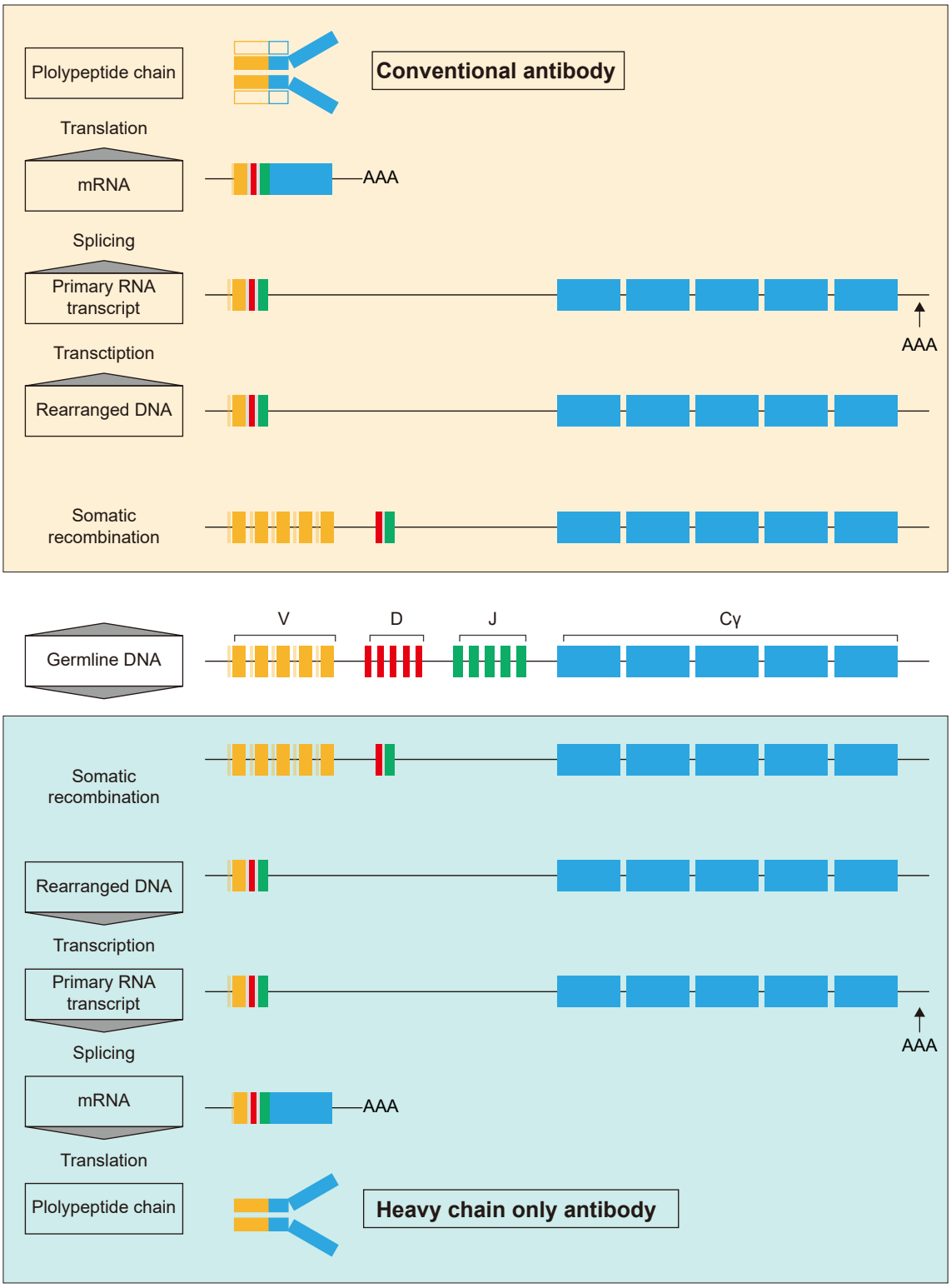
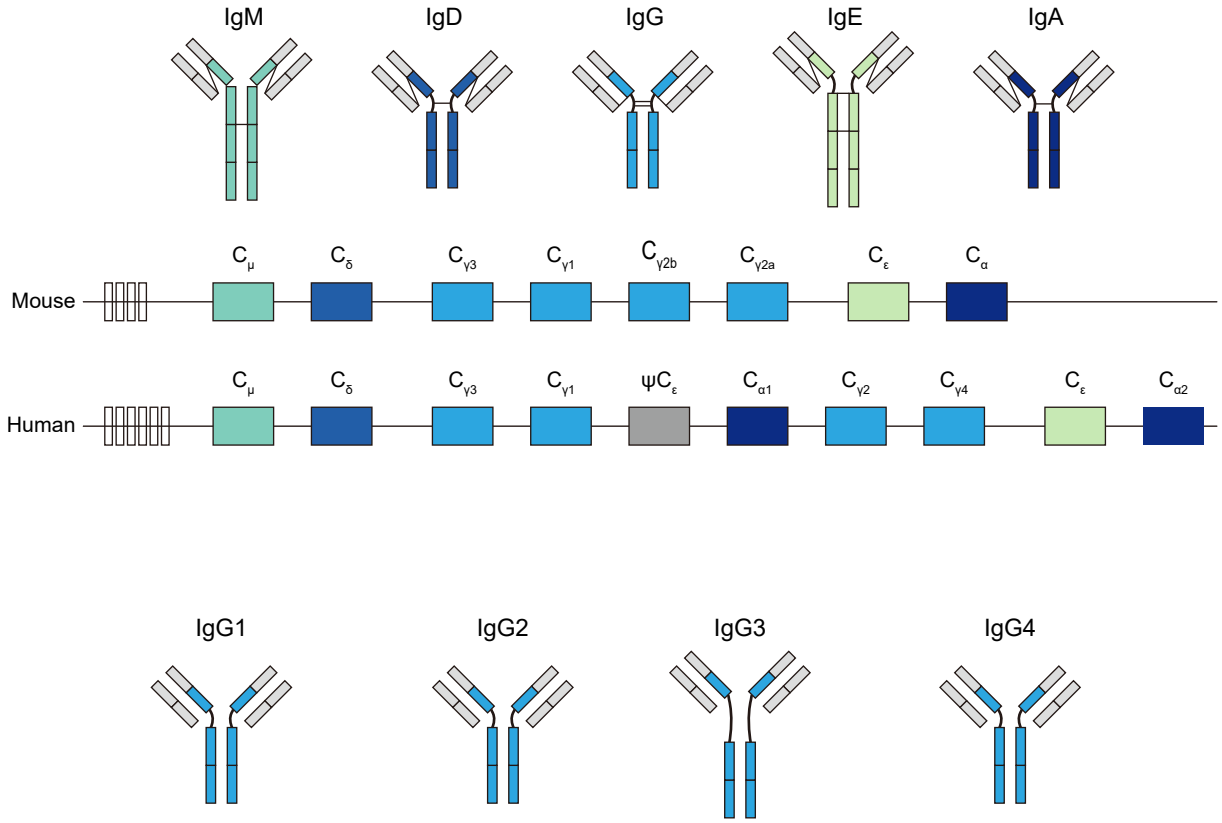

A
Antigen
Light chain (L)
Hinger region
Heavy chain (H)
VH
VL
CH1
CL
S-S
CH2
CH3
VHH
Conventional IgG antibody (IgG)
Camelid HCAb IgG
Nanobody
B
VH
FR1
CDR1
V42
G49
L50
W52
CDR2
FR3
CDR3
FR4
FR2
CH1
Hinge
CH2
CH3
Genome
H-locus
IGHV1
IGHV4
IGHV3
IGHV3H
D
J
IGHM
IGHG2
IGHG1
IGHGE
IGHA
n
VHH
F42
E49
R50
G52
C
H1
H2
H3
FR1
FR2
FR3
FR4


Plolypeptide chain
Conventional antibody
Translation
mRNA
AAA
Splicing
Primary RNA transcript
AAA
Transctiption
Rearranged DNA
Somatic recombination
Germline DNA
V
D
J
Cγ
Somatic recombination
Rearranged DNA
Transcription
Primary RNA transcript
AAA
Splicing
mRNA
AAA
Translation
Plolypeptide chain
Heavy chain only antibody


IgM
IgD
IgG
IgE
IgA
Mouse
Cμ
Cδ
Cγ3
Cγ1
Cγ2b
Cγ2a
Cε
Cα
Human
Cμ
Cδ
Cγ3
Cγ1
ψCε
Cα1
Cγ2
Cγ4
Cε
Cα2

IgG1
IgG2
IgG3
IgG4

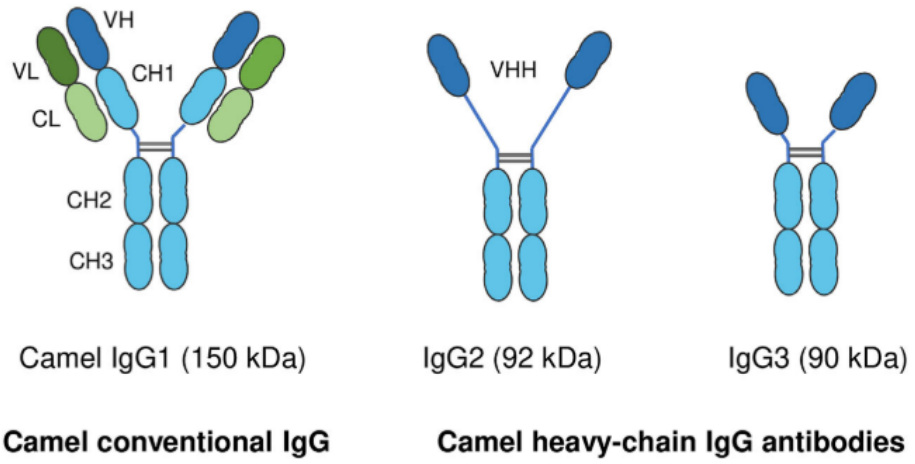


FIGURE 3 | Structure of camel immunoglobulin (Ig) G subclasses. Camel IgGs are currently classified into three structurally different isotypes: Camel IgG1 consists of two identical heavy chains (H) each composed of three constant domains (CH1–CH3) and a single variable domain (VH). Each heavy chain is covalently bound to identical light chains (L) with a constant (CL) and a variable domain (VL). Camel IgG2 and IgG3 are composed of only two identical heavy chains (long-hinge heavy chain in IgG2 and short-hinge heavy chain in IgG3). Camel single-chain IgG subclasses are devoid of the first heavy chain constant region CH1.

2a=35 3=12
2b=29
2c=15

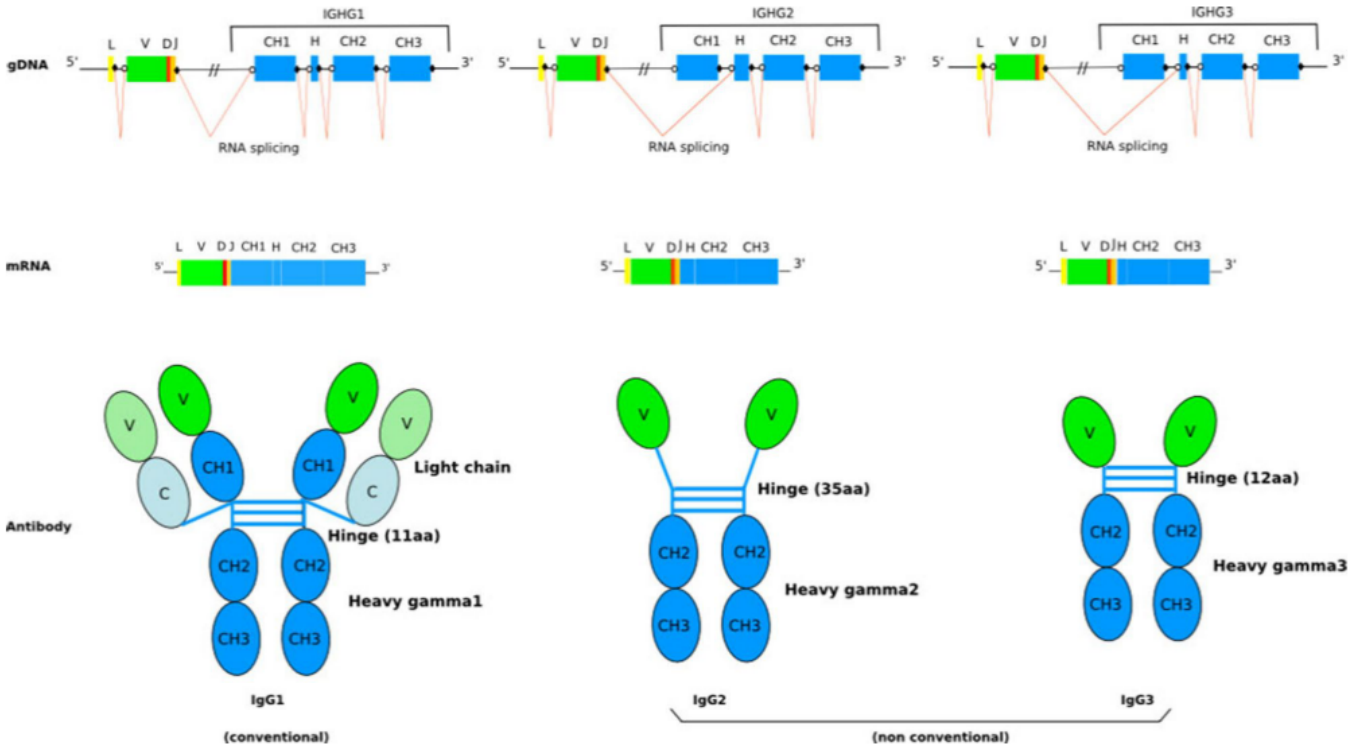


FIGURE 1 | Schematic representation of the general structure of camel conventional IgG1 and the nonconventional IgG2 and IgG3. The lack of light chain is itself a consequence of the absence of the CH1 domain in the gamma 2 and gamma 3 chains, due to a splicing defect of the CH1 exon. The figure is from IMGT: Camelidae IgG antibodies (http://www.imgt.org/IMGTbiotechnology/Camel_IgG.html#characteristics) with permission of IMGT®, the international ImMunoGeneTics information system ® (http://www.imgt.org).

Structural and genetic diversity in antibody repertoires from diverse species

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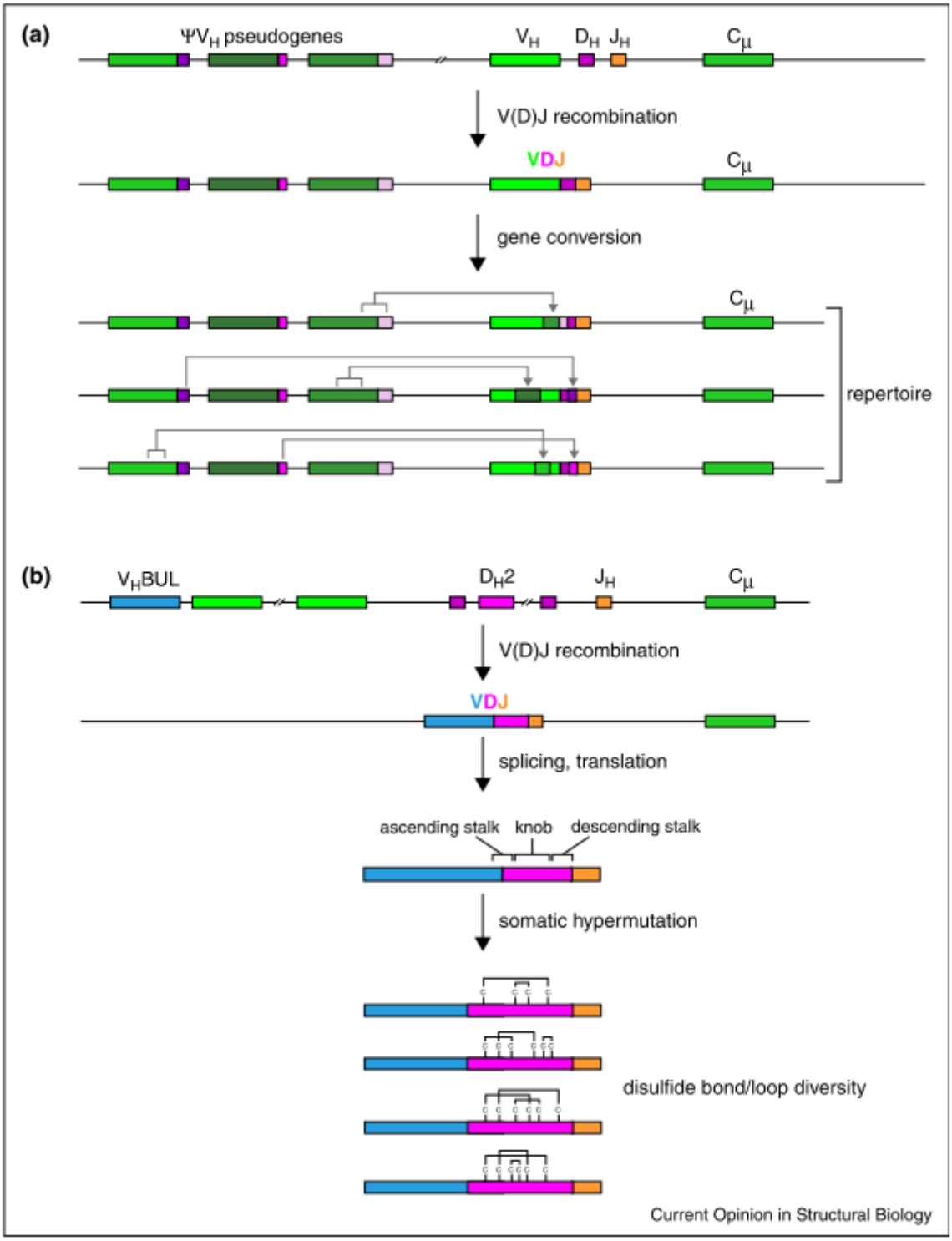
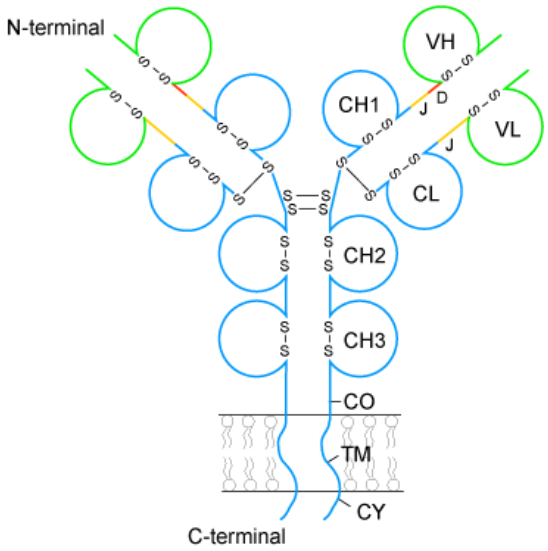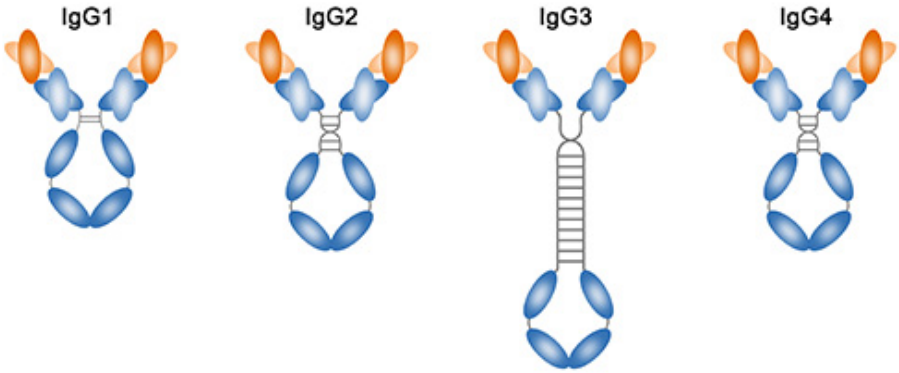
http://www.imgt.org/IMGTScientificChart/SequenceDescription/IG-GAMMA_KAPPA.png